

1.(a)

Under Assumptions 1-5, conditional on the values of explanatory variables

$$\text{Var}(\hat{\beta}_j | x_1, x_2, \dots, x_k) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ji} - \bar{x}_j)^2 (1 - R_j^2)} = \frac{\sigma^2}{SST_j (1 - R_j^2)}.$$

σ^2 : This is the variance of the error term, u , in the regression model. It represents the variability of the y variable that is not explained by the x variables.

x_{ji} : This denotes the value of the explanatory variable x_j for observation i

$\bar{x}_j$: This represents the mean of the independent variable x_j across all observations.

R_j^2 : it is the R-squared for the regression of x_j on all other explanatory variables. It measures the proportion of the variance in x_j that is explained by **all** the other x variables in the model.

(b) Discussed during the tut.

$\sigma^2, \sum_{i=1}^n (x_{1i} - \bar{x}_1)^2$ and R_1^2 affect the magnitude of variance.

$$(c) \text{sd}(\hat{\beta}_j) = \sqrt{\frac{\sigma^2}{SST_j (1 - R_j^2)}} \qquad se(\hat{\beta}_j) = \sqrt{\frac{\hat{\sigma}^2}{SST_j (1 - R_j^2)}}$$

(d) Since R_1^2 is expected to be large, we anticipate an inflated variance $\hat{\beta}_1$ (and standard error), consequently resulting in deflated t-statistic answer. For example, when we test $H_0: \beta_1 = 0$

against $H_1: \beta_1 \neq 0$, the t statistic answer is deflated, $t = \frac{\hat{\beta}_1 - 0}{se(\hat{\beta}_1)}$.

Similarly, we would have the same problem when testing β_2 , because a high R_2^2 also inflates the variance of $\hat{\beta}_2$

2. TRUE

Consistency and unbiasedness are related but distinct concepts.

Consistency: when the sample size, i.e., the number of observations, increases (tends to infinity), the estimator will converge to the true value of the parameter we are trying to estimate. In other words, it describes the long-term behavior of the estimator as we gather more data.

Unbiasedness: if we repeatedly collect samples and use the estimator to calculate the parameter each time, the average of those estimates will, over many repetitions, be equal to the true value of the parameter we are trying to estimate.

In other words, it describes on average, the estimator will give us the true value, no matter the sample size.